SPOTR™

RECON • DETECT • ALERT • OPTICAL PRECISION

Tactical Precision Camera Stream Viewer, Scope Reticle Customizer, First Focal Plane Engine, Decoupled Micro-Change Heatmap, Studio Voice Synthesizer, Dual In-Memory Buffer & WebHID Controller Virtual Mouse.

VERSION 0.1.1 SPECIFICATION | OFFLINE PWA READY | UVC & RTSP DUAL INGESTION

1080p @ 60 FPS	WebHID BT Mouse	RAM Ring Buffer	Studio Voice	100% Client-Side
UVC & RTSP GATEWAY	DUALSHOCK 4/5 TOUCHPAD & L2/R2	5S–60S INSTANT REPLAY	EMBEDDED AUDIO SPRITE TTS	ZERO EXTERNAL DEPENDENCIES

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OPTICS Dual Camera Ingestion: UVC USB & Wi-Fi RTSP

SPOTR features a universal dual video ingestion pipeline supporting both wired USB scopes and wireless network cameras.

- **UVC USB Scopes & Webcams:** Direct driverless HTML5 MediaDevices capture with automatic hot-plug renegotiation at 1080p/4K @ 60 FPS.
- **Bresser Full HD Wi-Fi Eyepiece Camera:** Integrated gateway (`server.py`) automatically dispatches UDP wake magic packets (`/api/rtsp/wake` to port 65534) and supports dual Anyka DSP streams:
  - **Channel 0 (1080p):** Full HD optics at `rtsp://192.168.1.1:554/live/ch0`.
  - **Channel 1 (540p):** High-framerate stream at `rtsp://192.168.1.1:554/live/ch1`.
- **Custom RTSP & ONVIF IP Cameras:** Connect any network IP camera via custom RTSP URL with sub-millisecond push-event frame distribution.
- **Synthetic Test Pattern:** Fallback test card with calibrated hash lines to verify compositor and reticle calibration without hardware attached.

> **CONDITION-PUSH FRAME DISTRIBUTION**
> The Python OpenCV gateway coordinates worker threads via `threading.Condition`, delivering incoming frames immediately to browser clients with zero polling jitter.

INPUT BUS DualShock 4/5 Wireless Controller & Virtual Mouse

Direct low-level USB/Bluetooth Human Interface Device integration utilizing high-speed **WebHID API** (Report ID 0x11).

- **Touchpad Virtual Mouse:** Clicking the center touchpad activates the high-contrast Virtual Mouse cursor. Gliding smoothly drives the pointer across all toolbars and sliders.
- **Tactical Triggers (L2 / R2):**
  - `L2 Trigger`: Left Mouse Click & Drag (selects buttons, scrubs timelines, drags resizable PiP loupe).
  - `R2 Trigger`: Right Mouse Context Click.
- **Precision Joysticks:** Left Stick controls primary scope reticle aim with a calibrated 50% precision rate curve. Right Stick smoothly pans the main feed when digitally zoomed.
- **7-Button Bluetooth Ring & Remote (1-to-1 Mode):** Hands-free finger ring control with ChromeOS HID swipe & zone calibration (`TouchTap@qxXqy`), live `RAW:` log, and 70% default pan speed in Setup → Controllers.

CONTROL MATRIX DualShock 4 & Remote Ring Command Reference

CONTROL INPUT	PRIMARY VIEWPORT ACTION	INSTANT REPLAY / GALLERY MODE	RETICLE CUSTOMIZER MODE
`Touchpad Click`	Toggle Virtual Mouse Cursor ON / OFF	Toggle Virtual Mouse Cursor ON / OFF	Toggle Virtual Mouse Cursor ON / OFF
`Touchpad Glide`	Precision Mouse Pointer Translation	Precision Mouse Pointer Translation	Precision Mouse Pointer Translation
`L2 Trigger`	Left Mouse Click / Drag Floating Windows	Left Mouse Click / Drag Timeline Scrubber	Left Mouse Click / Drag Slider Knobs
`R2 Trigger`	Right Mouse Click	Right Mouse Click	Right Mouse Click

CONTROL INPUT	PRIMARY VIEWPORT ACTION	INSTANT REPLAY / GALLERY MODE	RETICLE CUSTOMIZER MODE
`Left Joystick`	Primary Reticle Aim (Precision 50% Speed)	—	Navigate Menu Items (Up/Down) & Adjust Values (Left/Right)
`Right Joystick`	Viewport Pan (when Zoom > 1.0x)	—	Sculpt Concentric Rings & Center Gap Diameter
`L1 / R1 Bumpers`	Main Viewport Digital Zoom (− / +)	Playback Video Zoom (− / +)	Quick Step Click Indents (−1 / +1)
`D-Pad ▲ / ▼`	Reticle FFP Subtension Spacing (+5% / −5%)	—	Cycle Selected Setting / Drawer Focus
`D-Pad ◄ / ►`	Cyclic Quick-Tool Selector (Step Prev / Next)	Step Backward / Forward (1 Frame Forensic)	Decrease / Increase Active Value Slider
`Cross (✕)`	Toggle Active Cyclic Quick Tool	Play / Pause Video Playback	Toggle Numbered Subtension Hashmark Labels
`Circle (◯)`	Declutter Screen / Universal Back to Live	Close Replay & Return to Live Stream	Exit Customizer Screen
`Square (□)`	Toggle Reticle Overlay / Quick Drawer	Toggle 0.5x Slow-Motion Playback	Toggle Concentric Range Rings
`Triangle (△)`	Toggle Thermal Micro-Motion Heatmap	Toggle Saved Clips Drawer Panel	Toggle Center Bounding Brackets
`Share Button`	Instant Replay (Auto-Saves to RAM & IDB)	Export Native Video Clip (.webm)	Reset Reticle to Preset Defaults
`Options Button`	Open / Close Settings & Controller Screen	Exit to Live Stream	Exit to Live Stream
`Ring D-Pad (▲▼◄►)`	Pan Zoomed View (>1.0x, 70% default speed)	▲ Restart • ◄/► Step ±1 Frame • ▼ Return Live	Bindable in Setup → Controllers (HID Swipe / Key)
`Ring CENTER (●)`	Zoom Cycle (1x → 1.5x → 2x → 2.5x → 3x → 1x)	Play / Pause Toggle	Bindable in Setup → Controllers (TouchTap Zone)
`Ring BTN 2 / BTN 1`	Enter Instant Replay (BTN 2 Rec. on ChromeOS)	Exit Replay & Return to Live Stream	Bindable in Setup → Controllers (TouchTap Zone)

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VIEW 1 Live Feed

60 FPS live video, micro-change thermal heatmap, floating draggable PiP loupe, reticle drawer with FFP lock, and cyclic quick-tool dock.

VIEW 2 Instant Replay

Immediate native-resolution playback of the RAM buffer. Scrub frame-by-frame, auto-save to IndexedDB, or engage 0.5x slow-motion.

VIEW 3 Video Archive

Permanent local storage library in IndexedDB. Organize shot clips, delete with individual red trash icons, and export directly as .webm.

VIEW 4 & 5 Reticle & Controller

Parametric reticle studio (mils/MOA click corrections) & full-screen Gamepad & Bluetooth Ring setup with Learn schematic & RAW telemetry.

CONTROLS Floating Toolbars & Reticle Quick Drawer

- **Top-Left Control Strip:**
  - **Settings Gear:** Opens central configuration (camera source, RAM meter, controller mappings).
  - **Target POI Icon:** Toggles the resizable PiP Digital Zoom Loupe.
  - **Crosshair Reticle Icon:** Opens the Reticle Quick Drawer with live FFP toggle, opacity, reticle scale, and palette swatches.
  - **Thermal Heatmap Flame:** Toggles 20Hz micro-motion detection.
  - **Voice Announcement Speaker:** Quick-toggle for studio audio sprite speech engine.
- **Top-Left Zoom Pod:** Whole-viewport digital zoom controls (`-` , `1.0x` , `+` , `1:1`) with smooth touch and joystick panning.
- **1-Tap Declutter Mode:** Tapping anywhere on open screen space (or pressing `Circle`) hides all toolbars and buttons, leaving an uninhibited, 100% full-screen sight picture.

OPTICS Digital Zoom Loupe (#pipCanvas)

The Picture-in-Picture (PiP) Loupe provides real-time digital magnification of any specific point of interest (POI) on the target without changing the primary field of view.

- **Independent Pan & Zoom:** Adjust loupe magnification from 2.0x to 16.0x independently of main viewport zoom.
- **Sub-Pixel Target Crosshair:** A high-contrast target marker indicates the exact optical coordinate currently magnified in the loupe.
- **Draggable & Resizable:** Drag the loupe anywhere on screen by its header. Drag the corner handles to resize.
- **Hardware Steering:** Push the Right Joystick on your gamepad to smoothly track the loupe target across the target face.

APPEARANCE Tactical Color Palettes & Global Interface Opacity

Customize UI chrome and reticle colors across 6 tactical themes: **Emerald Teal** (#2dd4bf), **Amber Gold** (#ffd166), **Ruby Red** (#f87171), **Night Blue** (#60a5fa), **Amethyst** (#a855f7), or **High-Vis White** (#f8fafc).

Interface Opacity Slider (15% to 100%): Fades background chrome, borders, and buttons into near-invisibility (`--ui-opacity`), allowing spotting downrange without night-blindness or screen glare.

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DETECTION 20Hz Decoupled Micro-Change Heatmap Engine

SPOTR features an advanced micro-movement detection algorithm that highlights subtle changes (bullet impacts, target movement, wind drift) as a high-contrast thermal glow overlay.

- **20Hz Decoupled Pipeline:** Heatmap analysis executes on an independent 50ms timer, decoupling heavy frame diffing from the 60 FPS compositor to prevent UI hitching.
- **Downsampled Luma Analysis:** Evaluates a high-speed 160×90 luminance matrix (14,400 cells) against an exponential moving average background.
- **Global Shift Rejection (>18%):** Automatically discards whole-frame camera movements (wind buffeting, scope recoil, tripod bumps) to prevent false alerts.
- **Adjustable Sensitivity (10% to 95%):** Fine-tune detection thresholds in Settings to spot minute paper tears at 1000 yards without triggering on heat mirage.
- **Decay Duration (0.5s to 10s):** Controls how long the visual impact heat bloom persists before fading.

SPEECH Studio Voice Synthesizer & Web Speech Engine

When movement or bullet impact is detected, SPOTR computes the geometric centroid vector relative to your scope crosshair and vocalizes directional corrections.

- **High-Fidelity Studio Voice Sprite:** Uses Web Audio API (`voiceAudioEngine.js`) with studio-recorded female voice clips (`voice-sprite.mp3`) for crystal-clear, zero-lag announcements.
- **Browser TTS Fallback:** Activates native `SpeechSynthesisUtterance` if Web Audio is unsupported.
- **Clock & Reticle Click Announcements:** Announces target clock angle (1 to 12 o'clock) and distance in graduated reticle clicks: e.g. "*Detection! 11 o'clock, 3 clicks*".
- **Instant Voice Mute:** Click the speaker icon in the top-left toolbar or press the Mute hotkey to instantly silence audio without disabling visual heatmap rendering.

VOCABULARY Directional Speech Dictionary & Configuration Options

RELATIVE ANGLE	DEFAULT ANNOUNCEMENT	CUSTOM KEYWORD OPTION	OPERATIONAL USE CASE
337.5° – 22.5° (12 o'clock)	**"Target 12 o'clock"**	"North" / "Top" / "Elevation Up"	Impact high of point of aim; adjust elevation down.
22.5° – 67.5° (2 o'clock)	**"Target 2 o'clock"**	"High Right" / "North-East"	Impact high right; windage left + elevation down.
67.5° – 112.5° (3 o'clock)	**"Target 3 o'clock"**	"East" / "Right" / "Windage Right"	Crosswind deflection; adjust windage turret left.
112.5° – 157.5° (4 o'clock)	**"Target 4 o'clock"**	"Low Right" / "South-East"	Impact low right; hold over and into wind.
157.5° – 202.5° (6 o'clock)	**"Target 6 o'clock"**	"South" / "Bottom" / "Elevation Down"	Impact low; adjust elevation turret up.
Reticle Center (<5% radius)	**"Direct Hit" / "Bullseye"**	"Center" / "Hit"	Point of impact matches zero point of aim.

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IN-MEMORY Native-Resolution Circular RAM Ring Buffer

SPOTR maintains a **100% RAM-only rolling queue** (`memoryRingBuffer.js`) with zero disk writes during surveillance, protecting flash memory from wear.

- **Dual Source Ingestion:** Seamlessly ingests full-resolution frames from either UVC USB video or RTSP network streams (`getRtspCanvas()`).
- **Configurable Duration:** Store from **5 to 60 seconds** of rolling historical video in RAM.
- **Configurable Buffer FPS:** Set buffer capture between **5 and 30 FPS** to optimize RAM footprint on low-memory devices.
- **Live RAM Budget Meter:** Real-time telemetric bar in Settings displays Current Used MB, Estimated Allocation, and Free Device RAM.
- **Automatic Memory Guardrails:** Proactively purges older frames if device memory pressure exceeds threshold.

PLAYBACK Instant Replay Viewport & 0.5x Slow-Motion

Capture critical shot strings instantly with a single button press or ring tap.

- **Instant Snapshot:** Press `Share` on controller, press Ring `BTN 2` (or `BTN 1`), or click **REPLAY** to freeze and inspect the circular buffer.
- **Automatic Silent Persistence:** Upon trigger, the snapshot is automatically cloned to permanent local IndexedDB storage in the background without stuttering playback.
- **Precision Timeline Scrubber:** Drag the timeline slider to locate the exact microsecond of bullet impact or target change.
- **0.5x Slow-Motion Toggle:** Dedicated `0.5x` button allows smooth slow-motion trajectory and bullet impact forensics.
- **Dual-Direction Frame Stepping:** Step frame-by-frame forward (`STEP +1`) and backward (`STEP -1`) for precision ballistic analysis.

STORAGE Video Archive Library, Rapid Deletion & WebM Exporter

1. IndexedDB Local Storage

Clips are stored securely in browser sandboxed storage. Clips remain accessible across browser restarts and offline field sessions without cloud connectivity.

2. Rapid Per-Video Deletion

The collapsible Clips Drawer provides an individual red trash icon on every clip card, allowing rapid single-tap deletion without loading the full video player.

3. One-Click WebM Export

The native export engine encodes recorded frames into a standard `.webm` video file, saving it directly to local disk for reporting, archiving, or sharing.

RETICLE Parametric Reticle Studio & First Focal Plane (FFP)

Craft the ideal sighting overlay for any optics magnification or target geometry in View 4 (#screenReticle) or via the Quick Drawer:

- **First Focal Plane (FFP) Lock:** Dynamically scales hashmark spacing and reticle subtensions in lockstep with whole-image digital zoom, maintaining true angular mil/MOA values.
- **Extended 30-Click Elevation/Windage Ladder:** Display up to 30 clicks per arm with major graduation labels (every 5 or 10 clicks) across vertical and horizontal axes.
- **Optical Dead-Center Measurement:** All click distances and voice callout turret adjustments are calculated strictly from true optical dead-center (0, 0), completely independent of visual center gap diameter.
- **4 Selectable Center Aiming Symbols:**
  - **EMPTY:** Open unobstructed center viewing window.
  - **DOT:** Precision center aiming dot at the optical origin.
  - **CROSS:** Fine mini-crosshair intersection at dead-center.
  - **CHEVRON:** Tactical inverted caret (Λ) whose sharp apex points directly to the center pixel.
- **Direct Controller Sculpting & Cycle:** Adjust click count, spacing, gap, stroke weight, or press `Key C` / Quick Tool dock to cycle center point styles live.

CALIBRATION MOA & MRAD Turret Click Calculator

Eliminate sighting guesswork by calculating exact optical turret click offsets directly from observed bullet impacts:

- **Target Distance Input:** Enter engagement distance in Yards or Meters.
- **Turret Value Selection:** Select your scope turret click standard (1/4 MOA, 1/8 MOA, 1/2 MOA, 0.1 MRAD, or 0.05 MRAD).
- **Point of Impact (POI) vs Point of Aim (POA):** Measure target offset in millimeters or inches on the digital display.
- **Instant Turret Solution:** SPOTR calculates exact turret clicks (e.g. "*Elevation: +6 Clicks UP | Windage: -4 Clicks LEFT*") to achieve pinpoint zero.

DEPLOYMENT Progressive Web App (PWA) Offline Installation Guide

SPOTR is built as a self-contained, offline-first Progressive Web App. Precached by Service Worker `spotr-v0.1.1` (auto-reloading on update, with live build version pills in Setup headers):

Portable Windows (One-Click)

Double-click **start_spotr.bat**. Automatically boots the Python RTSP gateway and launches in your default browser.

ChromeOS / Chromebook

In Chrome, click the **Install SPOTR** icon in the omnibox. Runs as a standalone app window 100% offline.

Android Tablets & Phones

Tap browser menu (⋮) > **Install App / Add to Home Screen**. Operates in full-screen immersive landscape mode.

macOS & Linux

Run `python server.py 8080` and navigate to `http://localhost:8080` in Chrome, Edge, or Brave.